The ϕ^4 field as a second laboratory

The derivation, the estimator, the first measurements, and the experiments in flight

Jérôme's email of 1 October 2026 (after Marc) · this report 1 October 2026 · branch `claude/phi4-field-2026-10-01` · revised the same afternoon: reversibility, the physicist's parametrisation, E33's oracles in

Summary

Jérôme and Marc asked for a test of the estimator on Markov data from a physical model: spatial snapshots of the one-dimensional stochastic Allen–Cahn (ϕ^4) field, whose stationary law is a nearest-neighbour Gibbs measure, with four arms tracked across temperature and lattice spacing (a learned pair energy, the Gaussian closure, the window CNN, the exact score), and the suggestion that with this concrete example the paper may be complete. Everything in this report was built, tested and registered on 1 October; the deterministic part (E32) has run and is frozen; the learning part (E33, eight cluster jobs) and the physics part (E34) are in flight.

1. **The chain is exact and in the package** (§1). Its kernel is a Gaussian step tilted by a function of the *arrival* point, the column normaliser being the ground-state transform. The Gaussian AR(1) chain is the one member the two laboratories share (Proposition 3); the paper's mixture kernel is misspecified here through a nonlinear drift, not through its innovation shape.
2. **The learned pair energy is a transition kernel** (§2): concave M-step, the gradient is Boltzmann learning's positive phase minus a *conditional* negative phase, no chain-wide partition function (Proposition 6).
3. **The directed kernel and the symmetric pair energy are one model class** (Proposition 7, proved this afternoon): every tilted walk is reversible, with invariant law gZ and site energy $f = -\frac{1}{2}\log(g/Z)$, which for the field is the potential. A fitted kernel therefore reads as physics, $\hat\Theta = \hat\gamma s^2/h$ and $\hat V$ from $\hat f$ (Corollary 8); and the parsimonious exactly specified model parametrises f itself: the quartic site energy, five parameters, Jérôme's W_θ literally (§2.1). It joins E33 as a fifth arm under a registered amendment with two further predictions.
4. **One thing in the email does not survive** (§3): moment projection of the messages is not a well-posed estimator on this chain, because the backward message is nearly flat. Measured: 42–100 % of a projected message's mass at the grid edge. The Gaussian-closure arm is therefore the covariance twin, which is what Proposition 1 of the paper proves projection computes on linear chains.
5. **E32** (§4): at matched correlation $\lambda_1/\lambda_0 = 0.85$ the linear floor is 0.269 at $\Theta = 0.3$ against 0.168 for the Laplace laboratory, and falls to 0.041 at $\Theta = 2$; the operating grid is exact to 10^{-12}; Theorem 2's gap ratio rises monotonically through t_* to 0.73–0.94 at $t = 3$.
6. **E33 and E34** (§5): five kernel arms through exact BP, the window head and the transformer under both interfaces, the oracles; the physical statistics the consolidated note asks for; the clean-data interface. Eleven predictions were registered before any confirmatory cell ran. The oracles have landed: P6 is confirmed on the per-seed test bundles (floor 0.269, 0.114, 0.059, 0.041 along $\Theta = 0.3, 0.5, 1, 2$; sixteen seeds, spread below 0.001). The kernel and network jobs are queued; a laptop preview on two cells and two seeds is running.

Notation. Θ is the physical temperature (the code and the cell names call it `T`), h the lattice spacing, t the diffusion time of the channel $x_t = e^{-t}a + \sqrt{1-e^{-2t}}\,z$, L the number of sites, M the number of training sequences, N_g the grid size. The consolidated note's σ is our $\sqrt{1-e^{-2t}}/e^{-t}$.

1 The chain

1.1 From the SPDE to a Markov chain

The equation $\partial_s\phi = \partial_x^2\phi - V'(\phi) + \sqrt{2\Theta}\,\eta$ with $V(\phi) = (\phi^2-1)^2/4$ is the gradient flow of $H[\phi] = \int[\frac{1}{2}(\partial_x\phi)^2 + V(\phi)]\,\mathrm{d}x$ at temperature Θ; in one space dimension its stationary law is the Gibbs measure $\propto \exp(-H/\Theta)$. On a lattice of spacing h, $H_h(\phi) = \sum_i[(\phi_{i+1}-\phi_i)^2/2h + hV(\phi_i)]$, a nearest-neighbour energy. A spatial snapshot is therefore a first-order Markov chain in the site index. Writing the site terms symmetrically,

$$\mathsf{T}(\phi,\phi') = D(\phi)\,G(\phi'-\phi)\,D(\phi'), \qquad D = e^{-hV/2\Theta}, \quad G(u) = e^{-u^2/2\Theta h}, \tag{1}$$

is a symmetric, positive kernel, Hilbert–Schmidt for a confining V, with a simple largest eigenvalue λ_0 and a positive eigenfunction ψ_0, $\int\psi_0^2 = 1$.

Proposition 1 (ground-state transform)**.** *$K(\phi' \mid \phi) = \mathsf{T}(\phi,\phi')\psi_0(\phi')/(\lambda_0\psi_0(\phi))$ is a Markov kernel; $\pi = \psi_0^2$ is invariant for it; and the chain is reversible, $\pi(\phi)K(\phi' \mid \phi) = \pi(\phi')K(\phi \mid \phi')$.*

Proof. $\int K(\phi' \mid \phi)\,\mathrm{d}\phi' = \int \mathsf{T}(\phi,\phi')\psi_0(\phi')\,\mathrm{d}\phi'/(\lambda_0\psi_0(\phi)) = 1$ by the eigen-equation. $\int \pi(\phi)K(\phi' \mid \phi)\,\mathrm{d}\phi = \psi_0(\phi')\int \mathsf{T}(\phi,\phi')\psi_0(\phi)\,\mathrm{d}\phi/\lambda_0 = \psi_0(\phi')^2$ by symmetry of T and the eigen-equation again. $\pi(\phi)K(\phi' \mid \phi) = \psi_0(\phi)\mathsf{T}(\phi,\phi')\psi_0(\phi')/\lambda_0$ is symmetric in (ϕ,ϕ'). □

The infinite-volume Gibbs measure restricted to L consecutive sites is this stationary chain, $\pi(\phi_1)\prod K(\phi_{i+1} \mid \phi_i)$; that is what "spatial snapshots of a stationary field" means, and it is the law the data are drawn from. (A finite open chain with free ends is also Markov but inhomogeneous within a correlation length of its ends; we do not use it.) The lag-k covariance is $\sum_{n\geq 1}(\lambda_n/\lambda_0)^k\langle\psi_0|\phi|\psi_n\rangle^2$, so $\rho_1 := \lambda_1/\lambda_0$ plays the role ρ plays for the AR(1) family and the correlation length in sites is $1/\log(1/\rho_1)$.

1.2 The tilted-walk identity

Proposition 2 (tilted walk)**.** *With $g := D\psi_0$,*

$$K(\phi' \mid \phi) = \frac{G(\phi'-\phi)\,g(\phi')}{Z(\phi)}, \qquad Z(\phi) = \int G(\phi'-\phi)g(\phi')\,\mathrm{d}\phi' = \frac{\lambda_0\psi_0(\phi)}{D(\phi)}. \tag{2}$$

Proof. From (1), $\mathsf{T}(\phi,\phi')\psi_0(\phi') = D(\phi)G(\phi'-\phi)g(\phi')$, and the eigen-equation reads $D(\phi)\int G(\phi'-\phi)g(\phi')\,\mathrm{d}\phi' = \lambda_0\psi_0(\phi)$, which is the stated Z. Substituting in Proposition 1's K gives (2). □

So the kernel is a Gaussian random-walk step of variance Θh, tilted by a positive function of the arrival point and normalised per column: Jérôme's "Gaussian increment tilted by $\psi_0(\phi')e^{-hV(\phi')/\Theta}/\psi_0(\phi)$", with the normaliser made explicit. The point for the estimator is that *the column normaliser is the ground-state transform*: a family that parametrises $\log g$ and normalises columns on the grid contains the truth exactly, and never needs ψ_0 or λ_0.

1.3 Where the two laboratories meet

Proposition 3 (Gaussian embedding)**.** *The Gaussian AR(1) kernel $\mathcal{N}(a';\rho a,q)$ with $0<\rho<1$ is the member of (2) with step variance $\gamma = q/\rho$ and $\log g(a') = -(1-\rho)a'^2/2q + \text{const}$. No other kernel of innovation form $p_\varepsilon(a'-\rho a)$, $\rho\neq 0$, belongs to the family, and no other member of the family has a state-independent innovation.*

Proof. $\log\mathcal{N}(a';\rho a,q) = -a'^2/2q + \rho a a'/q - \rho^2 a^2/2q + \mathrm{c}$ and $\log[G_\gamma(a'-a)g(a')] = -a'^2/2\gamma + aa'/\gamma - a^2/2\gamma + \log g(a') + \mathrm{c}$. Matching the cross term gives $\gamma = q/\rho$; the a'^2 terms give $\log g(a') = -a'^2(1/q-\rho/q)/2$, a Gaussian of variance $q/(1-\rho) = 1+\rho$; the a^2 terms are absorbed by $Z(a)$. For the second claim, $\partial_a\partial_{a'}\log K$ equals $1/\gamma$, a constant, for every member of (2), and equals $-\rho(\log p_\varepsilon)''(a'-\rho a)$ for an innovation kernel, which is constant iff $\log p_\varepsilon$ is quadratic. □

Remark 4 (the misspecification mechanism). Expanding $\log g$ around a, $K(a' \mid a) \approx \mathcal{N}\big(a';\, a+\gamma(\log g)'(a),\,\gamma\big)$: at fine h the chain is the Euler step of $\mathrm{d}a = \gamma(\log g)'(a)\,\mathrm{d}s + \sqrt{2\gamma}\,\mathrm{d}W$ with a *nonlinear* drift. The mixture-innovation kernel $\sum_c \pi_c\mathcal{N}(a'-\rho a;\mu_c,s_c^2)$ has a linear drift and a state-independent innovation; it is misspecified here through the drift. That is a different mechanism from every axis of the paper's misspecification study (global latent, long-range coupling, AR(2), site-dependent ρ, heteroskedasticity), and E33's P1/P2 measure it.

1.4 Standardisation, cells

Everything downstream assumes unit marginal variance, so the package exposes $a = \phi/s$, $s^2 = \mathrm{Var}_\pi(\phi)$ (the mean vanishes by symmetry); $K_a(a' \mid a) = sK_\phi(sa' \mid sa)$. The (Θ,h) map (Fig. 1) gives the iso-correlation line $h^*(\Theta)$ at $\rho_1 = 0.85$, the laboratory's frozen correlation: it passes through $\Theta = 0.30$ at $h = 1$. The temperature sweep sits on that line, so the second laboratory is compared with the first at matched second-order structure while the marginal kurtosis runs from -1.35 to -0.46; the refinement sweep halves h twice from the base cell with $L \propto 1/h$, so the physical length is fixed and the correlation length in sites grows $6 \to 12 \to 24$ (Table 1). The step width $\sqrt{\Theta h}/s$ is at least 14 grid cells on every cell, so the usual resolution condition is met with room.

cell	Θ	h	L	ρ_1	ξ (sites)	s	$\kappa_4(\pi)$
base	0.3	1	32	0.848	6.08	0.821	−1.352
T-sweep	0.5	0.419	32	0.850	6.15	0.766	−0.881
T-sweep	1.0	0.226	32	0.850	6.15	0.818	−0.588
T-sweep	2.0	0.150	32	0.850	6.15	0.950	−0.461
refine	0.3	0.5	64	0.917	11.47	0.805	−1.253
refine	0.3	0.25	128	0.957	22.48	0.800	−1.218

Table 1: The six confirmatory cells (`outputs/E32/cells.json`). h^* solved by bisection to $\rho_1 = 0.85 \pm 10^{-4}$; $\xi = 1/\log(1/\rho_1)$.

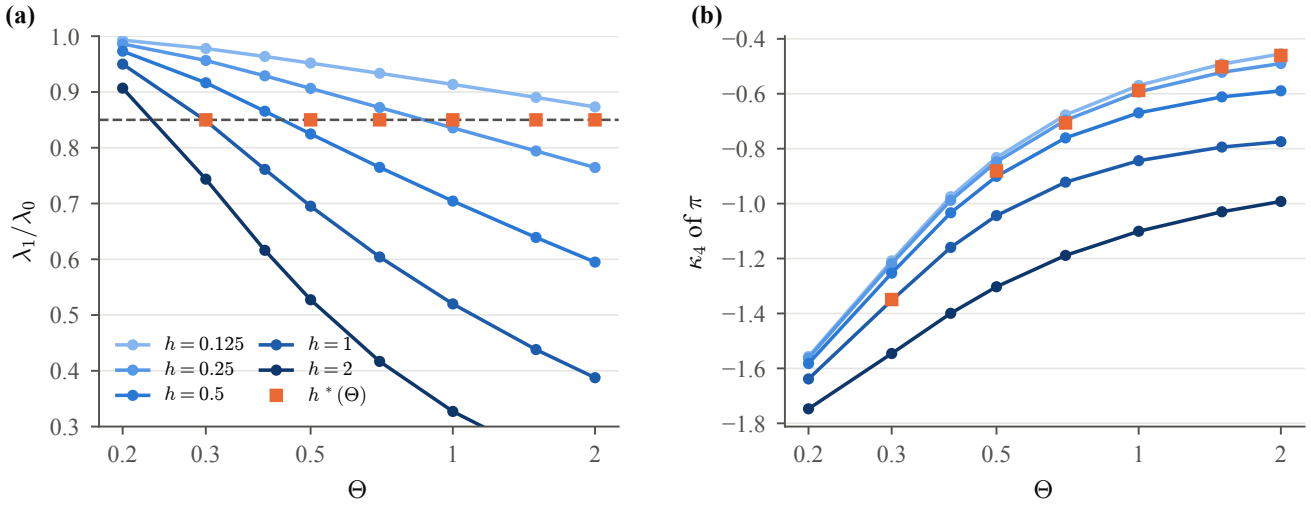


Figure 1: The (Θ, h) map of the chain. (a) λ_1/λ_0 against Θ for five spacings; the dashed line is the laboratory's $\rho = 0.85$ and the squares are the solved $h^*(\Theta)$. (b) Excess kurtosis of the standardised marginal: the non-Gaussianity knob, platykurtic throughout (bimodal), from -1.75 to -0.46.

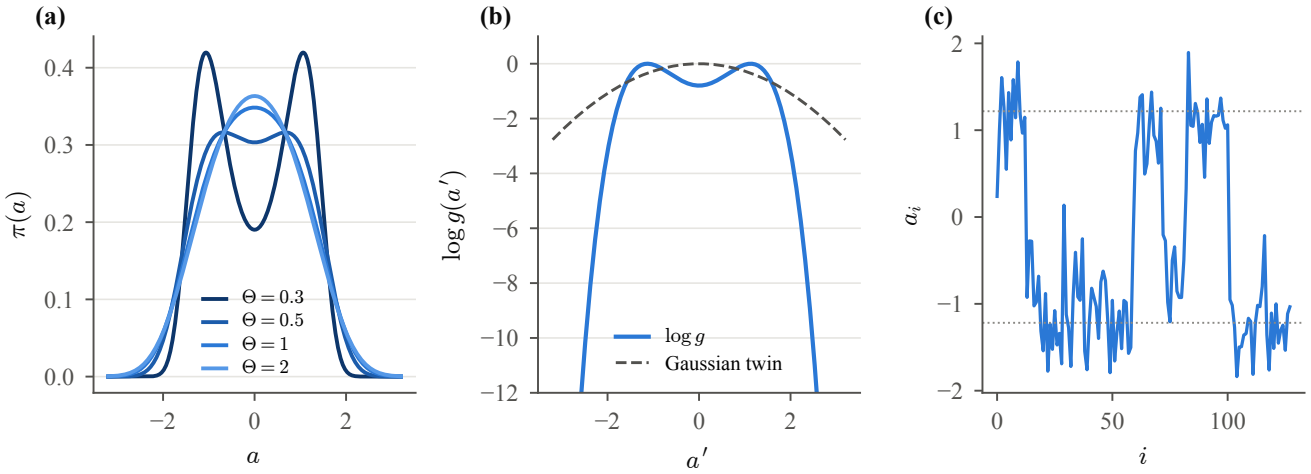


Figure 2: The chain at the temperature cells. (a) The standardised stationary law $\pi = \psi_0^2$: two wells at $\pm 1/s$ merging as Θ rises. (b) The arrival-point tilt $\log g$ at the base cell against the quadratic tilt of its Gaussian twin (Proposition 3): the truth is bimodal and super-Gaussian, the twin is a parabola. (c) One exact sample of 128 sites at the base cell; dotted lines at the wells. Domains of 5–25 sites separated by kinks.

2 The estimator: the learned pair energy as a kernel

The family is $K_\theta(a' \mid a) = \mathcal{N}(a'; a, \gamma)\exp(B(a')\theta)/Z_\theta(a)$ with B a linear basis for $\log g$ (Legendre polynomials on $[-A, A]$, or cubic B-splines), the constant function omitted (it is absorbed by Z), and Z the column normaliser under the grid's trapezoid weights, exactly as the MDN kernel is normalised. Parameters $(\log\gamma, \theta)$. In Jérôme's language, the pair energy is $W_\theta(a, a') = (a' - a)^2/2\gamma - \log g_\theta(a')$ and the transfer-matrix normaliser is $Z_\theta(a)$.

Lemma 5 (the M-step is concave in the tilt)**.** *With Ξ_{kj} the expected transition mass on $u_j \to u_k$ from the E-step and $N_j = \sum_k \Xi_{kj}$, $Q(\theta) = \langle \Xi, \log K_\theta \rangle = \sum_k (\sum_j \Xi_{kj}) B_k\theta - \sum_j N_j \log \sum_k w_k e^{B_k\theta + c_{kj}}$, linear minus a sum of log-sum-exp functions, hence concave in θ at fixed γ; its gradient is $\nabla_\theta Q = B^\top \sum_j \tilde{\Xi}_{\cdot j}$ with $\tilde{\Xi}_{kj} = \Xi_{kj} - N_j p_j(k)$, p_j the model's column distribution, and $\partial Q/\partial \log\gamma = \sum_{kj} \tilde{\Xi}_{kj}[(u_k - u_j)^2/2\gamma - \frac{1}{2}]$.*

Proposition 6 (what Boltzmann learning becomes for a kernel)**.** *By Fisher's identity the exact gradient of the noisy-data log-likelihood is $\langle \Xi, \nabla \log K_\theta \rangle$, which for this family is*

$$\nabla\mathcal{L} = \sum_j \Big[\sum_k \Xi_{kj}\, \nabla(-W_\theta(u_j, u_k)) - N_j\, \mathbb{E}_{K_\theta(\cdot\mid u_j)}\big[\nabla(-W_\theta(u_j, \cdot))\big] \Big] :$$

the positive phase is the posterior pair expectation (as in the note's §16.2) and the negative phase is the kernel's own conditional *expectation at each source state, weighted by the posterior source mass. The undirected pair-energy*

model of the note's §16 has instead the chain's equilibrium *pair marginal as negative phase and a chain-wide* $\log Z_\theta$*. Both are exact maximum likelihood for their model; the directed form needs no* Z_θ*, its column normalisers are the ground-state transform (Proposition 2), and it contains the field's kernel exactly.*

Proposition 7 (every tilted walk is reversible)**.** *Let* $\gamma > 0$ *and* $g > 0$ *with* $Z_g(a) = \int \mathcal{N}(a'; a, \gamma) g(a')\, \mathrm{d}a' < \infty$*, and let* $K_g(a' \mid a) = \mathcal{N}(a'; a, \gamma) g(a')/Z_g(a)$*. Then* $\pi_g \propto g Z_g$ *is invariant and* K_g *is reversible with respect to it. Equivalently* K_g *is the ground-state transform of the symmetric operator* $\mathsf{T}_g(a, a') = u(a)\mathcal{N}(a'; a, \gamma)u(a')$ *with* $u = \sqrt{g/Z_g}$*, whose Perron eigenfunction is* $\psi = \sqrt{gZ_g}$ *with eigenvalue* 1*; so* K_g *is the Gibbs chain of the pair energy*

$$W(a, a') = \frac{(a' - a)^2}{2\gamma} + f(a) + f(a'), \qquad f = -\tfrac{1}{2}\log(g/Z_g), \tag{3}$$

and conversely every pair energy of this form is a tilted walk. For the field's (γ, g) *of* (2)*,* $f(a) = (h/2\Theta)V(sa) +$ const*.*

Proof. $\pi_g(a)K_g(a' \mid a) \propto g(a)g(a')\mathcal{N}(a'; a, \gamma)$ is symmetric in (a, a'): detailed balance, and invariance by integrating over a. For the second form, $(\mathsf{T}_g\psi)(a) = u(a)\int \mathcal{N}(a'; a, \gamma)u(a')\psi(a')\,\mathrm{d}a' = u(a)\int \mathcal{N}(a'; a, \gamma)g(a')\,\mathrm{d}a' = u(a)Z_g(a) = \psi(a)$ since $u\psi = g$; and $\mathsf{T}_g(a, a')\psi(a')/\psi(a) = \mathcal{N}(a'; a, \gamma)\, g(a')/Z_g(a)$ since $u\psi = g$ and $\psi/u = Z_g$. The converse is Proposition 2 with $D = e^{-f}$. For the field, $Z_g = \lambda_0\psi_0/D$ by (2), so $g/Z_g = D^2/\lambda_0$ and $f = -\log D + \text{const} = (h/2\Theta)V + \text{const}$. □

Numerically, on the $(401, 8)$ grid: π_g by the formula and by an eigensolve of K_g agree to 4×10^{-12}, detailed balance holds to 10^{-14}, and the site energy of the field's own kernel equals $(h/2\Theta)V(sa)$ to 2×10^{-10} (`tests/test_pair_energy_kernel.py`). The first question put to Jérôme and Marc in the morning version of this report (directed kernel or symmetric pair energy?) dissolves: no tilted walk is irreversible, and the symmetric form has the same single free function. What differs is the algorithm, per-column normalisers against the Perron eigenvalue.

Corollary 8 (a fitted kernel reads as physics)**.** *In standardised units* $\gamma = \Theta h/s^2$ *and* $f(a) = (h/2\Theta)V(sa) + \text{const}$*. A fit* $(\hat\gamma, \hat g)$ *of any kernel of this class gives*

$$\hat\Theta = \hat\gamma s^2/h, \qquad \hat V(\phi) = \frac{2\hat\Theta}{h}\,\hat f(\phi/s) + \text{const}, \qquad \hat f = -\tfrac{1}{2}\log\left(\hat g/Z_{\hat g}\right) :$$

the temperature from the bond variance, the one parameter that enters no tilt; the potential, its barrier and its wells, from the site energy. Moreover the conditional mean step is exactly $\gamma(\log Z_g)'(a)$ *(exponential tilting), which for the field is* $\Theta h(\log\psi_0)' + (h^2/2)V'$ *in physical units, so the small-h limit is* $\mathrm{d}\phi = \Theta(\log\psi_0)'\,\mathrm{d}x + \sqrt{\Theta}\,\mathrm{d}B$*: the drift of the stationary field in the site index is the log-derivative of the ground state, not* $-V'$*.*

2.1 The physicist's parametrisation

Since $\log g = \log\psi_f - f$ is a transcendental function of the potential while f itself is a quartic, the parsimonious exactly specified model parametrises f. `PairEnergyKernel` (`src/markov_score/pair_energy_kernel.py`): $f(a) = \sum_{m \le 4} c_m (a/2)^m$ and $\log\gamma$, five parameters, with the same fixed soft wall as an energy; K_f by one dense eigensolve of T_f per evaluation (15 ms on 401 points), invariant law ψ_f^2 in closed form; the M-step maximises $Q = \langle \Xi, \log K_f \rangle$ by L-BFGS-B with the analytic gradient of the Perron eigenvector from first-order perturbation theory in the complete discretised eigenbasis, $\partial_{c_m}\psi_0 = -\sum_{n \ge 1} \psi_n \frac{\lambda_0 + \lambda_n}{\lambda_0 - \lambda_n}\langle \psi_n \mid B_m\psi_0 \rangle$, checked against finite differences and Fisher's identity to 10^{-4}. Q is not concave in c, so a step is accepted only if Q did not decrease, as for every other kernel. With the field's own (γ, c) the kernel reproduces the chain's to 2×10^{-3} (the coarse eigensolve against the fine one), and from 1024 clean chains at $(\Theta, h) = (0.3, 1)$ it reads $\hat\Theta$ within 5 % and the barrier within 10 %. This is literally the pair energy W_θ of the email, its Boltzmann learning written as a kernel. It joins E33 as a fifth arm, `em_pair_quartic`, under amendment 1 of the registry (predictions P7 and P8, declared before any E33 kernel result existed), run as its own shards so the four registered arms' tables are never touched. In the smoke run at $M = 32$ it already sits at 0.13 against the polynomial tilt's 0.25: five parameters have the smaller variance.

Implementation (`src/markov_score/tilted_kernel.py`): L-BFGS-B on $(\log\gamma, \theta)$ with the analytic gradient, warm-started, accepted only if Q did not decrease (a rejected step returns the receiver's parameters in a new object, so the checkpoint ladder stays honest); Fisher's identity checked against finite differences of the evidence to 10^{-5}; the Gaussian embedding recovered on a Gaussian chain, and the field's kernel recovered from clean data with the spline basis (`tests/test_tilted_kernel.py`). Two details earned by the first smoke run:

- **The tilt carries a fixed soft wall** $-(a/5)^{12}$ on the $(401, 8)$ grid. The data occupy $|a| < 4$, so the basis beyond that is constrained only through far columns' normalisers, and a free fit can raise the tilt at the grid edge and create a metastable pocket there, invisible to the likelihood but the place where the kernel's invariant law then puts mass. The wall is 0.002 at $|a| = 3$, 0.07 at 4 and 280 at 8, more than any coefficient within the optimiser's bounds can undo; the field's true tilt is already $-0.2a^4$ there.

- **The initial law is the kernel's own invariant density**, by a direct eigensolve of the 401×401 operator (`em.fit_em` takes a callable `log_mu`): the model is a stationary chain, as the data are. The M-step maximises the edge term; any evidence decrease is recorded.

3 What does not extend: Gaussian closure of the messages

The paper's Proposition 1 identifies moment-projected sum-product with the LMMSE of the covariance twin, and its proof uses the linear transition. On the field chain the projected recursion is not a well-posed estimator, for a reason that was found by the diagnostic, not anticipated.

Proposition 9 (the backward message saturates)**.** *Let* $\log g(a') \le -c\,a'^4 + C$ *(a super-Gaussian tilt,* $c > 0$*). Then* $K(\cdot \mid a)$ *concentrates, as* $|a| \to \infty$*, near* $a'_*(a)$ *with* $|a'_*| \sim (|a|/4c\gamma)^{1/3}$*, and the backward message* $R(a) = \int K(a' \mid a)\ell(a')\,\mathrm{d}a' = \mathbb{E}[\ell(a_{i+1}) \mid a_i = a]$ *is bounded and decays only as* $\exp(-c'|a|^{2/3})$*.*

Sketch. The exponent $-(a'-a)^2/2\gamma - ca'^4$ is maximised where $(a-a')/\gamma = 4ca'^3$, i.e. $a'_* \approx (a/4c\gamma)^{1/3}$ for large $|a|$, with curvature $\sim 12ca'^2_*$, so $K(\cdot \mid a)$ is a sharpening Gaussian at a'_* and $R(a) \approx \ell(a'_*(a))$; for a Gaussian ℓ of width $\sqrt{\Delta}/\alpha$ this is $\exp(-\alpha^2(a'_* - x/\alpha)^2/2\Delta)$. □

So on a finite grid R is nearly constant, and the Gaussian with its moments is close to flat: the projection discards the backward information. Measured at $t = 0.47$ on the $(401, 8)$ grid, 28 % of a projected message's mass sits in the outer 5 % of the grid; over E32's cells and levels the maximum is 42 % at the base cell and 98–100 % at every other cell (Table 2). On grids wide enough for the Gaussian and Laplace chains' backward messages (half-width 32 at $t = 1$) the projected recursion reproduces the LMMSE to 10^{-8} and 10^{-5}, so this is a property of the field, not of the implementation (`tests/test_bp_projected.py`).

Consequently the Gaussian-closure arm of the field chain is its covariance twin, the LMMSE of the true covariance, by the identification Proposition 1 proves on linear chains; the projected recursion is kept as the diagnostic and its edge mass is recorded. Jérôme's expectation that closure "fails badly" by smearing across domain walls is then measured by the linear floor, which is what the floor has always measured, and it does: 0.269 at the base cell. For the paper this is a scope sentence on Proposition 1 and one sentence in the field section.

4 E32: the laboratory before any learning

Deterministic, registered and predeclared, run on the laptop at commit `9b1c940`, frozen (`outputs/E32`, 9 files). Three predictions were registered first.

cell	linear floor	isotropic	projected (diag.)	edge mass	grid error
$\Theta = 0.3$, $h = 1$, $L = 32$	0.269	0.674	0.262	0.42	9.6×10^{-15}
$\Theta = 0.5$, h^*, $L = 32$	0.115	0.658	0.278	1.00	1.3×10^{-12}
$\Theta = 1$, h^*, $L = 32$	0.058	0.657	0.208	0.99	1.3×10^{-15}
$\Theta = 2$, h^*, $L = 32$	0.041	0.656	0.169	0.98	1.6×10^{-15}
$\Theta = 0.3$, $h = 0.5$, $L = 64$	0.162	0.714	0.291	1.00	2.3×10^{-15}
$\Theta = 0.3$, $h = 0.25$, $L = 128$	0.098	0.750	0.529	1.00	1.1×10^{-15}

Table 2: E32 ladder, pooled relative score error over the twelve-level schedule (mean of two replicates of 512 chains), the projected recursion's largest edge mass over levels, and the worst relative error of the $(401, 8)$ exact score against $(1601, 8)$ over levels. For scale: the Laplace laboratory's floor at $\rho = 0.85$ is 0.168 and E14b's bimodal chain's 0.082.

P1, the grid: confirmed far inside its bound. The $(401, 8)$ exact score agrees with the $(1601, 8)$ one to 10^{-12} relative or better on every cell and level, against a declared tolerance of 10^{-3}. The field kernel is smooth, so the trapezoid rule is spectrally accurate here, unlike the Laplace chain's kink. The same holds for the eigensolve: halving the fine spacing or widening the domain moves λ_0, ρ_1 and s by less than 10^{-10}.

P2, the projection: confirmed. The edge mass exceeds 10 % at some level of every cell (Table 2); the closure arm is the covariance twin.

P3, Theorem 2: consistent, not reached within the schedule. The ratio rises monotonically through t_* and reads 0.85, 0.87, 0.73, 0.94 at $t = 3$ on the four cells (Fig. 4). The AR(1) families reached the constant at $t = 4$ with 8192 chains (E18); this run stops at $t = 3$ with 512, and the ratio at $t = 2$ already moves by 0.1 between the two replicates. The constants are large, $C_4 = \frac{1}{6}\|\kappa_4\|_F^2 = 3882$, 2350, 1261, 838 at $\Theta = 0.3, 0.5, 1, 2$ for $L = 32$, which is why the gap is still 10^{-2}–10^{-1} of the score energy at $t = 1$. Extending the schedule to $t = 4$ with 8192 chains is cheap and is the one addition E32 needs.

The floor itself. At the base cell the linear floor is 0.269, well above the Laplace laboratory's 0.168 at the same correlation and more than three times E14b's bimodal chain's 0.082 (kurtosis −0.72): the field at $\Theta = 0.3$

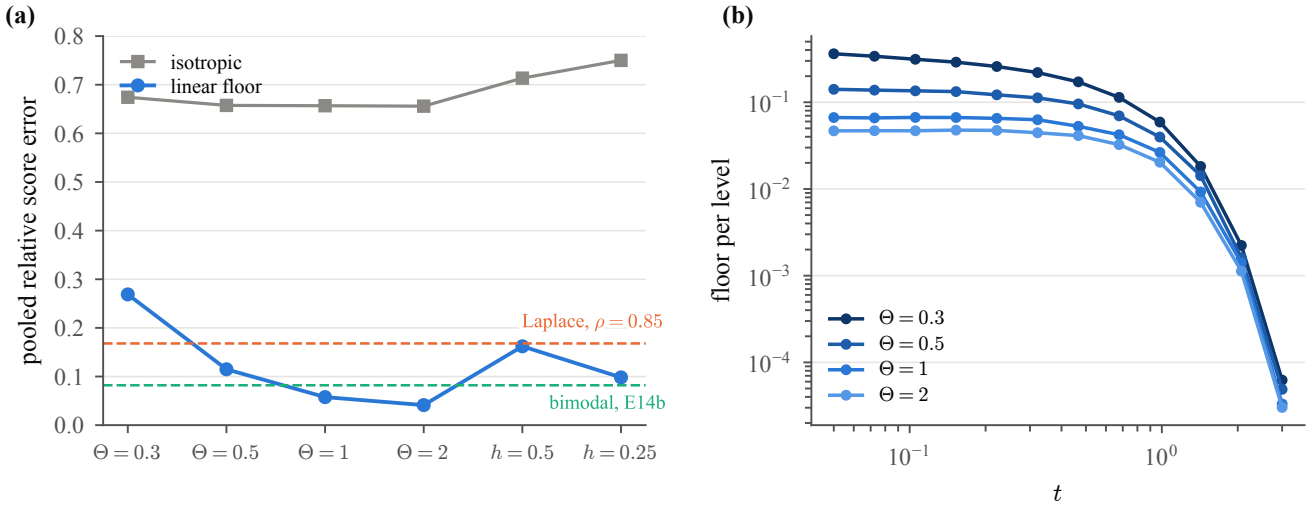


Figure 3: (a) The linear floor and the isotropic map across the six cells; dashed references are the laboratory's Laplace floor and E14b's bimodal floor at the same correlation. (b) The floor per level at the four temperature cells: the field's second-order deficit lives at low t and is largest where the wells are deepest.

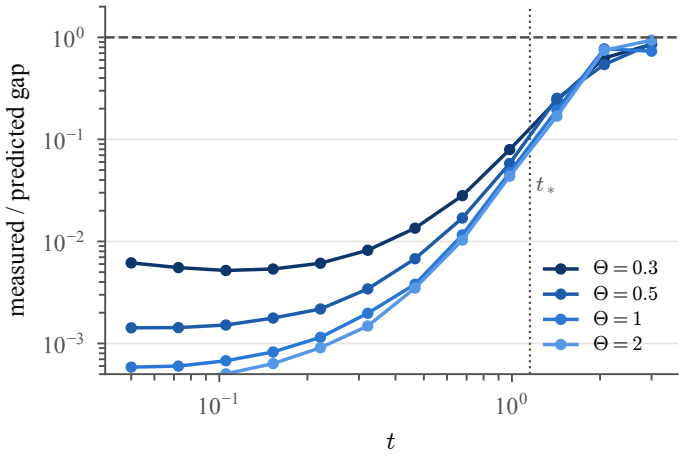


Figure 4: Theorem 2 on the second laboratory: $\mathbb{E}\|S - S^{\mathrm{G}}\|^2$ from tree-exact BP against the linear floor, divided by $\frac{1}{6}\|\kappa_4\|_F^2 e^{-8t}$ with κ_4 from transfer-matrix four-point functions, at the four temperature cells; $t_* = \frac{1}{2}\log\lambda_{\max}(\Sigma_0) = 1.15$.

carries more structure beyond second order than any chain the paper has used, and the exact score has that much room above the floor. The floor falls monotonically along the temperature sweep ($0.269 \to 0.041$), which is E33's P6, and along the refinement cells ($0.269 \to 0.162 \to 0.098$), because at finer h the per-step kernel is closer to Gaussian while the chain is more correlated.

5 E33 and E34: what is in flight

5.1 E33, learning through noise

On the six cells, under the registered protocol (sizes $M \in \{32, 128, 512, 2048\}$, seeds 0–15, bundles keyed by tag, seed, cell and M, one noised view per sequence spread over the twelve levels, validation 1024 and test 2048 chains per seed):

- kernel arms through exact BP, the same ladder to 1200 for every kernel, the kernel's own invariant law as initial law, both of E14's selection rules (held-out BP risk; held-out one-view evidence): `em_mixture` ($C = 8$, 25 parameters, misspecified), `em_tilted_poly` (Legendre degree 6, 7), `em_tilted_spline` (24 B-splines, 25), `em_mdn` (E25/E30's neural kernel, 1,516), and, by amendment 1, `em_pair_quartic` (the site energy itself, 5; §2.1);
- the window head and the transformer, re-screened per cell on development seeds and trained two ways, imported unchanged from E14: implicit score matching on the one-view bundle (the matched interface) and denoising score matching on clean targets with one fixed corruption (the network-favourable bracket), at $M \in \{2048, 128\}$;

- per (cell, seed): the linear floor, the isotropic map, the projected diagnostic.

Registered predictions. P1 the mixture keeps an asymptotic bias on every cell (above the spline in 16/16 seeds at $M = 2048$, the gap not closing from 512 to 2048). P2 the learned pair energy beats the mixture at every $M \geq 128$ on every cell, the 7-parameter tilt within 20 % of the 25-parameter one at $M = 2048$. P3 the MDN improves relative to the mixture compared with E30 (mdn/mixture below 1.5 at $M = 2048$): a state-dependent mean is what this chain needs. P4 no network reaches the spline at any M on any cell under either interface. P5 the window head's deficit grows along the refinement cells while the spline's does not. P6 the floor at the base cell is above E14b's 0.082 and falls along the sweep. Amendment 1 (afternoon, before any kernel result): P7 the five-parameter site energy is within 10 % of the spline at $M = 2048$ on every cell and below it at $M = 32$ and 128 (16/16 paired); P8 $\hat{\Theta}$ is within 10 % of Θ at $M = 2048$ for both the quartic and the spline (16/16), and the quartic's barrier within 20 % of 1/4 on the $\Theta \leq 1$ cells.

Oracles, in. Job 678084 completed after 2 h 04 and was fetched: 288 rows, sixteen seeds on all six cells. P6 is *confirmed* on the test bundles: the linear floor is 0.269 at the base cell (E14b's bimodal floor at the same correlation is 0.082) and falls monotonically along the sweep, 0.269, 0.114, 0.059, 0.041 at $\Theta = 0.3, 0.5, 1, 2$, with a spread over seeds below 0.001; on the refinement cells 0.164 and 0.098 at $h = 0.5, 0.25$. The projected diagnostic carries 38–43 % of its mass at the grid edge on the base cell and 98–100 % on every other cell, which is the measured form of §3.

Jobs, deployment `9b1c940-20261001T084745Z`, submitted 08:55 UTC: kernels 678087–678090 (`defq`, four seed shards, each its own directory); screen 678091 then nets 678092/678094 (`long_gpuh200`, one H200 each; the A100 partition was in maintenance); all still queued at the time of writing. The kernel shards are estimated at about eight hours each on 24 workers, the $L = 128$ cells at $M = 2048$ being the long pole; the network jobs at one to two days. The quartic arm's four shards are submitted from the afternoon's deployment (§2.1).

5.2 E34, the physics and the clean-data interface

The consolidated note (§24.5, §16.1) asks for physical statistics beside the score error and for the clean-data baseline. E34 supplies them on E33's streams, so every row pairs with E33's: `physics_truth` (kink density per bond and per unit length, interior domain lengths, $C(r)$ against the spectral sum, kurtosis against the exact one, barrier mass, on 4096 exact chains per cell; laptop); `clean` (the four kernel arms fitted by clean-data maximum likelihood on E33's training chains, scored on E33's test bundles: the price of the channel per family; one cluster job); `physics_learned` (ancestral samples from every fitted E33 kernel at $M = 2048$, started from its own invariant law, the same statistics against the truth; after E33). Four predictions registered; the quick smoke already shows the direction of two of them: on clean data at $M = 32$ the mixture sits at 0.27 while the polynomial tilt reaches 0.04, and samples from a fitted mixture overshoot the truth's kink density (0.21 against 0.13 per bond) while the spline's match it.

5.3 First readings, from a laptop preview

While the cluster shards queue, a preview of the kernel arms runs on the laptop: the base cell and the $\Theta = 2$ cell, seeds 0 and 1, $M \in \{128, 512\}$, under the registered code path into separate directories (not confirmatory, two seeds). The five-parameter site energy finished first. Figure 5 and Table 3 are Corollary 8 applied to its risk-selected fits at $M = 512$, i.e. to kernels learned *through the noise* from 512 sequences of 32 sites each seen at one level.

Table 3: Readings from the risk-selected fits (preview; mean over the seeds, spread in brackets). Truth: $\hat{\Theta}/\Theta = 1$, barrier 1/4, wells at ± 1. "RMS" is the root-mean-square deviation of $\hat{V}$ from V on $|\phi| \leq 1.6$ after removing the constant.

cell	arm, data	M	$\hat{\Theta}/\Theta$	barrier	well	RMS	risk
$\Theta = 0.3$	site energy, noisy	128	0.980 (0.084)	0.256 (0.033)	1.02	0.019	0.035
$\Theta = 0.3$	site energy, noisy	512	1.005 (0.023)	0.234 (0.016)	1.00	0.016	0.021
$\Theta = 0.3$	site energy, clean	512	0.985 (0.010)	0.236 (0.002)	1.00	0.009	0.010
$\Theta = 0.3$	site energy, clean	2048	0.999 (0.005)	0.250 (0.000)	1.00	0.003	0.005
$\Theta = 0.3$	spline tilt, clean	2048	0.999 (0.005)	0.250 (0.003)	1.00	0.004	0.006
$\Theta = 2$	site energy, noisy	512	1.025 (0.056)	0.50 (0.33)	1.04	0.13	0.019
$\Theta = 2$	site energy, clean	2048	1.003 (0.004)	0.29 (0.02)	1.04	0.03	0.004

Three readings. First, the temperature is identified to about a percent and the double well to 0.016 RMS from noised data at $M = 512$, with a score risk of 0.021 against the Gaussian closure's 0.269 on the same cell: a factor thirteen, by a kernel with five parameters. At $M = 128$ the temperature is still read within 2 % on average (spread 8 %) and the barrier within 3 %; clean data at $M = 2048$ read all three to 0.1–0.5 %, so the channel costs a factor of a few in risk and a few percent in the physics at these sizes, not the identification. The same fitted

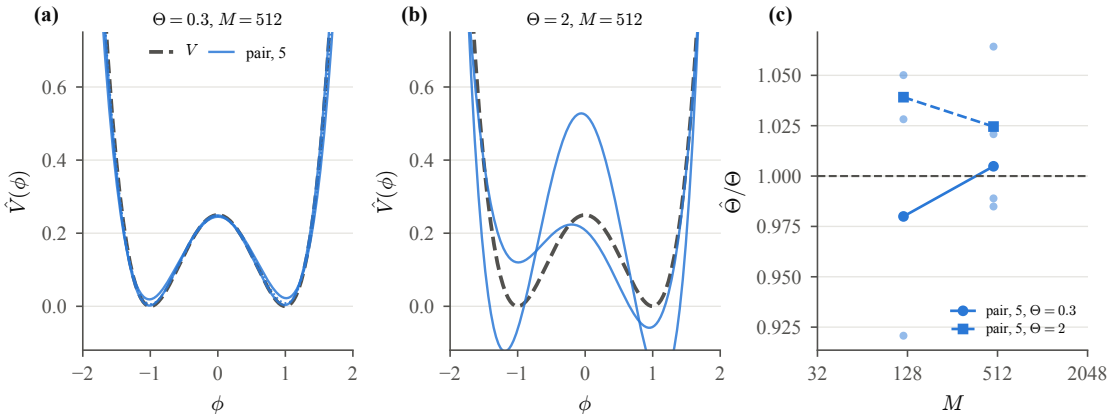


Figure 5: The potential and the temperature read from kernels learned through noise (preview, two seeds). (a) $\hat{V}(\phi)$ from `em_pair_quartic` at the base cell, $M = 512$, each seed a thin line, the true double well dashed, a clean-data fit dotted. (b) The same at $\Theta = 2$. (c) $\hat{\Theta}/\Theta$ against M. Regenerated by `tools/make_field_potential_figs.py`; the confirmatory shards replace the preview when they land.

kernels, sampled ancestrally from their own invariant law (E34 `physics_learned` on the preview), reproduce the truth's kink density per bond (0.123 and 0.130 against 0.123 and 0.120 on the two seeds) and its mean domain length (5.1 and 4.9 sites against 5.1 and 5.0): the generated-sample quality the note asks for, in the note's own statistics. Second, at $\Theta = 2$ the temperature is still read to 0.3–2.5 % but the barrier is not: 0.50 ± 0.33 through noise, 0.29 ± 0.02 even from clean data. At that temperature the thermal energy is eight times the barrier and the marginal is near-Gaussian, so the data carry almost no information about the shape of V between the wells; the risk is nonetheless 0.019 against the closure's 0.041, because what the score needs there is the bond and the curvature, which are identified. P8 was written to expect this. Third, the clean-data fits read everything to 0.5 % at the base cell: the price of the channel at $M = 2048$ is a factor of a few in risk, not in the physics.

One negative result came with the preview (N-33 in `docs/NEGATIVE_RESULTS.md`): the *polynomial* tilt fitted by clean maximum likelihood is ill-posed outside the data support. A counting-measure Ξ puts no mass beyond $|a| \approx 3.5$, the likelihood rises as $\log g \to -\infty$ there, and a global basis bends inside the bulk when pushed down outside it; on seed 1 the clean fit of the degree-6 Legendre tilt reached a risk of 1.79 while the spline gave 0.007 and the site energy is immune by construction. The role of the parsimonious exactly specified kernel passes to the site energy; the polynomial tilt's registered rows are read with an at-bound flag.

6 Alignment with the consolidated note

the note asks for	where it is
four arms across Θ, h, σ	E33: learned pair energy (two capacities), closure = covariance twin (§3), window head, exact score; twelve levels t; six cells
exact transfer-matrix sampling	`field.py`: eigensolve on 2401 points, continuum rejection sampler and lattice inverse-CDF sampler, agreeing
score error, posterior-mean error	E33's estimand; the posterior-mean error is the same quantity per level up to α/Δ
$C(r)$, ξ, kinks, domains, bimodality	E34 `physics_truth` and `physics_learned`
generated-sample quality	E34 `physics_learned`; the direct sampler of E27
runtime	E24x (uncontended reruns, in queue); the field adds nothing to BP's cost, which depends on N_g, L, B only
clean-data learning (§16.1)	E34 `clean`
symmetric W_θ, reversibility	by construction: every tilted walk is reversible (Proposition 7); the symmetric site energy is learned directly by `em_pair_quartic` (§2.1)
Θ read from the data	Corollary 8: $\hat{\Theta} = \hat{\gamma} s^2/h$ from any kernel of the class; P8
Fourier BP for the SPDE (§23)	not pursued: E24 found BP memory-bound (23.8 % of a sweep is the matmul), so a spectral step caps at 1.3×
"minimal decisive experiment"	E32 (floor across Θ, h) plus E33's window arm
figures-first (§28)	figs. 5 and 6 of the note are E32/E33's; the summary figure is pending the E33 data

7 Questions for Jérôme and Marc

1. The morning's question (directed kernel or symmetric pair energy?) is answered by Proposition 7: they are one class. The remaining choice is the *form* of the pair energy. The Gaussian-bond-plus-site-energy form (3) is the whole class of discretised gradient SPDEs, and the quartic site energy is exactly specified for ϕ^4

with five parameters; a general symmetric $W(a, a')$ would also cover non-gradient couplings at the price of a two-dimensional function to learn. Is that generality wanted for the paper, or is the site-energy form the right level, with the tilt bases as the agnostic alternative?

2. Gaussian closure of the messages is not well-posed on the field (§3); the closure arm is the covariance twin. Does that match what Marc had in mind by "Gaussian closure" for this benchmark, or did he mean the Kalman smoother of the linearised dynamics?

3. The temperature sweep is at matched $\lambda_1/\lambda_0 = 0.85$ rather than at fixed h, so that the field is compared with the AR(1) laboratory at matched second-order structure. Agreed, or would a fixed-h sweep be more natural for the physics reading?

4. Theorem 2's ratio is still 0.75–0.95 at $t = 3$: the ϕ^4 constants are large. Is it worth a sentence in the paper, or only the figure?

A Numerics

Eigensolve. $A = W^{1/2}\mathsf{T}W^{1/2}$ on 2401 points of $[-12, 12]$ (physical units), symmetric, `scipy.linalg.eigh`; $\psi_n = W^{-1/2}v_n$. Components below 10^{-13} of the maximum are eigensolver noise and are carried as $\log\psi_0 = -\infty$ (stationary mass below 10^{-26}); a kernel column leaving such a state is the normalised T column. Spectral accuracy: $|\Delta\lambda_0|/\lambda_0 < 10^{-10}$, $|\Delta\rho_1| < 10^{-10}$ under halving the spacing and under widening the domain to $[-16, 16]$. Completeness: $\sum_n c_n^2 = \mathrm{Var}_\pi(\phi)$ to 10^{-10}.

Samplers. Continuum: propose $\phi' \sim \mathcal{N}(\phi, \Theta h)$, accept with $g(\phi')/\max g$, $\log\psi_0$ by cubic spline on the resolved range; burn-in $\lceil \log(10^{-12})/\log\rho_1 \rceil$ steps from $\mathcal{N}(0, s^2)$. Lattice: inverse-CDF on the columns of the discretised kernel. Agreement in variance, lag-one correlation and kurtosis within Monte Carlo error at 3000 chains.

Fourth cumulant. $\mathbb{E}[a_i a_j a_k a_l]$ for $i \le j \le k \le l$ equals $\langle\psi_0|\, a\, R^{j-i} a\, R^{k-j} a\, R^{l-k} a\,|\psi_0\rangle$ with $R = \mathsf{T}/\lambda_0$, a triple sum over the leading 256 modes with $\Phi_{nm} = \langle\psi_n|a|\psi_m\rangle$; κ_4 by Wick subtraction. The diagonal reproduces the marginal excess kurtosis to 10^{-6}.

B Tests

33 new tests (59 pass in the files they live in, with the pre-existing ones): `test_field.py` (14: stationarity, reversibility, column mass, the tilted-walk identity, refinement, widening, completeness, two samplers, cumulant diagonal, BP as a denoiser, the iso-correlation solver), `test_tilted_kernel.py` (7: the Gaussian embedding, Fisher's identity by finite differences, M-step ascent and contract, clean MLE on the Gaussian chain and on the field, EM through noise), `test_bp_projected.py` (6: identity and LMMSE on wide grids for the Gaussian and Laplace chains, the truncation failure pinned, the field's near-flat backward message pinned), `test_stationary.py` (+2: eigensolve against power iteration; the wall), the kernel M-step contract (+4). The fast suite is unchanged (178 passed).